



Lightweighting Process of Digital Twin Information Models for Smart City Services

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ABSTRACT

A smart city digital twin refers to a system designed to digitally replicate real-world city objects in a virtual space and implement real-time synchronization with the real world, thus helping to solve urban problems through various service scenario simulations and data analysis processes. In relation to the development of city information models for implementing the smart city digital twin, many efforts have been made to integrate and utilize BIM-GIS data. However, due to problems with the capacity available for storing large amounts of data, there are limitations in practical applications because it is sometimes difficult to provide numerous real-time services in the era of web environments. Therefore, in this study, a lightweighting process for the Lightweight Digital Twin System was presented as a method to provide various services that a smart city offers based on the digital twin, and a lightweighting method was proposed accordingly. The process and method were verified through the development of a case application for the case of 00 city. It is expected that the Digital Twin System lightweighting method proposed in this study will serve as a basis for efficient city information processing and contribute to facilitating the provision of various smart city-related services.

1. Introduction

The Smart City is an urban management system capable of improving the quality of life for its citizens while solving urban problems. Globally, the Smart City has been increasingly introduced in various fields such as energy, transportation, and healthcare (Caragliu et al., 2011; Wan et al., 2019). With a growing interest in the Smart City, the development of corresponding technologies has been required to address various urban problems, and Digital Twin technology has been in the spotlight for the efficient operation of Smart City based on vast amounts of data (Tao et al., 2018; Shahat et al., 2021).

The Digital Twin is a virtual representation of urban space in the real world, and it can support problem solving phases via simulation in virtual space and analysis processes (Wan et al., 2019; Lim et al., 2020). Meanwhile, the concept of space is essential to establish a digital twin system for the Smart City, and its establishment should be based on the spatial information of

real-world city components. Recently, there have been efforts to develop smart city services using urban object information in the macroscopic aspect of urban facility units, such as roads, bridges, tunnels, and buildings with LOD (Level of Detail) 3 (Karimi and Iordanova, 2021) or lower based on 2D GIS information or GIS standards (e.g., CityGML, City Geographic Markup Language), including disaster management-related research regarding the improvement of situational awareness and decision-making (Kim et al., 2019), and healthcare service-related research to improve mobility for the elderly through a linkage between wearable biosensors and a Geographic Information System (GIS) (Lee et al., 2020), etc. However, for services such as energy management and facility management, object information on the facility interior, that is, Building Information Modeling (BIM) information from a microscopic point of view is needed (Lee et al., 2021).

Therefore, GIS that displays geographic information from a macroscopic perspective and BIM that contains detailed information about buildings/urban facilities from a microscopic point of view

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are needed to establish a digital twin at the city level for general-purpose smart city services, and many efforts have been made to integrate and utilize them as urban management systems (Rich and Davis, 2010; Mignard and Nicolle, 2014; Kang and Hong, 2015; Park and Kim, 2016; Pishdad-Bozorgi et al., 2018). However, models built at the current LOD 3 level are only available for geometric visualization (Ruohomaki et al., 2018; Schrotter and Hürzeler, 2020), and although there are studies designed to convert BIM object information into GIS surface information, additional work for a physics simulation is needed for the surface model. In addition, because BIM standard-based information utilization or visualization using a Web 3D engine has a complex structure and contains a lot of parametric information due to the characteristics of the information model, it is difficult to create a digital twin at the city level. In recent years, global efforts have been made to specify ISO/TS 19166:2021 (BIM to GIS conceptual mapping) international standards (ISO/TS 19166, 2021) for BIM-GIS data integration. However, schema integration or mapping between two heterogeneous models is very complex and can cause a variety of ambiguity problems. Moreover, it is still in its infancy like conceptual mapping, and physical schema integration increases data capacity even if it is successful. Therefore, separate work for lightweighting is needed for the digital twin model to provide various services in Smart City (Wang et al., 2021).

In this regard, this study seeks to present a process and methods to create a lightweight digital twin information model

as a basic requirement to provide various services in a Smart City based on the digital twin. Lightweighting methods for the digital twin system can be divided into a method of reducing the capacity of a 3D geometric model from system components, a method of selectively extracting and displaying information and a method of visualization through texture adjustment in terms of system operation. This study focused on the method of making the model lightweight and selectively extracting and displaying information.

2. Literature Review

2.1 Digital Twin Research

The concept of the digital twin began with a virtual prototype used for simulation within the virtual space facility environment of NASA's Apollo program and has emerged as a key element technology of the 4th industrial revolution as it is actively applied in the manufacturing industry. As shown in Table 1, various digital twin-related studies have been conducted recently. Examples include semi-automatic generation of city digital twin models, and smart city digital twin architecture etc. Some studies suggest the concept of a lightweight digital twin for the purpose of flexible manufacturing and MES (Manufacturing Execution System) framework (Liu et al., 2022). However, those studies attempted to reduce the operational workload of the digital twin by improving the product manufacturing simulation process (Zhang et al., 2021; Zhang et al., 2022). Also, in terms of service provision,

Table 1. Literature on Digital Twin

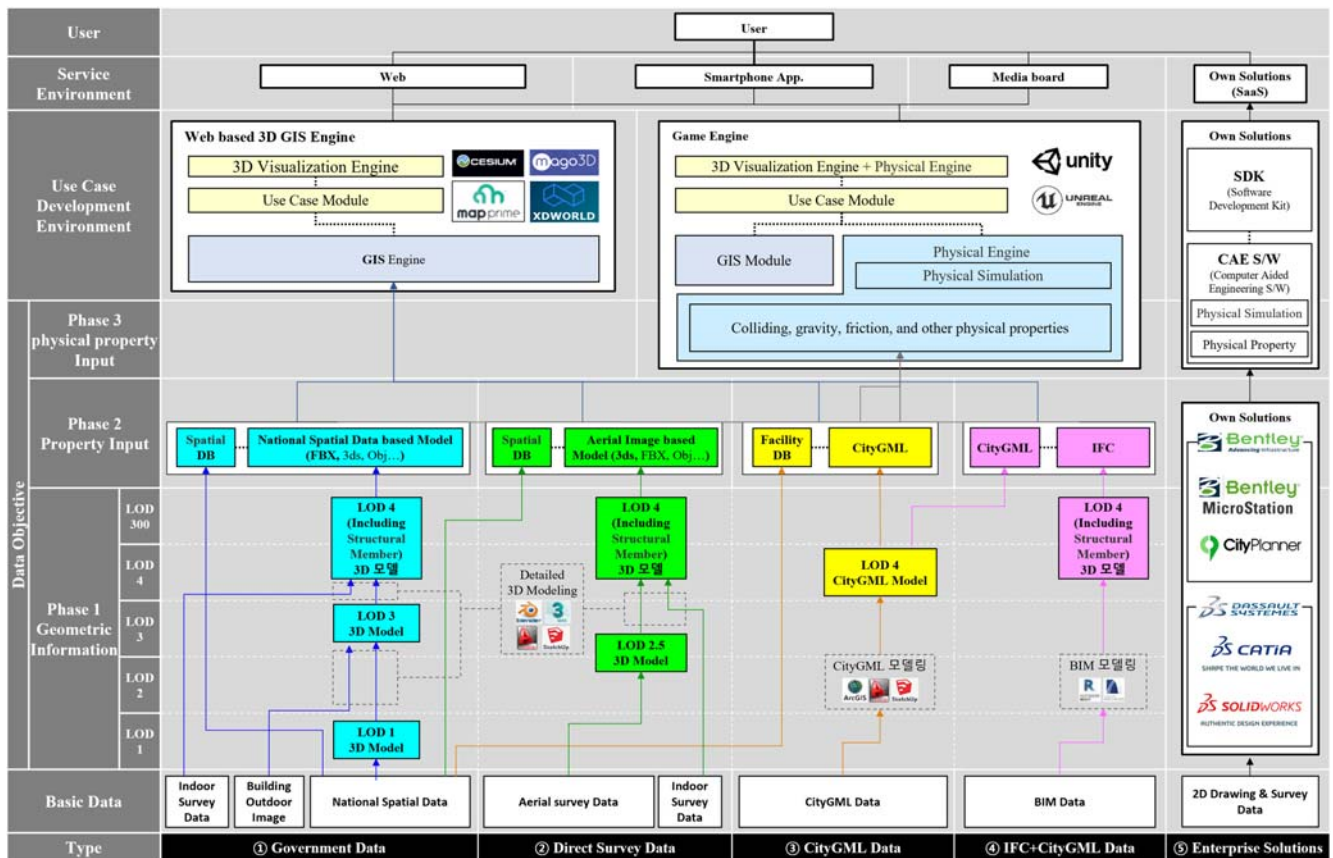
References	Advantages	Limitations	Lightweighting
Xue et al., 2020	City object recognition from LiDAR point clouds using the cross-sections and taxonomy unsupervised method	The approach cannot recognize incomplete or auxiliary geometries scanned from LiDAR point clouds	No
Lu et al., 2020	Semi-autonomous geometrical digital twinning approach using images and CAD drawings	The approach can be implemented at the level of LOD300 (AIA)	No
Lehner and Dorffner, 2020	The study Proposes the idea of a digital geo twin concept to create all city information model by defining detailed LOD	It is an idea level, and there is a limit to the level of accuracy for each LOD	No
Fang et al., 2021	The study presents a vision for the digital twin paradigm for interdisciplinary convergence in the fields of ICT and AI	The study points out the limitations of AI related to data labeling, rumor detection, to reduce the dimensionality and redundancy of multimodal data	No
Fang et al., 2021	The study proposes a method for reducing the weight of various types of complex CAD models for production line equipment	It is a method of reducing the shape information of the CAD model and does not deal with the property information	Yes
Deren et al., 2021	It is an architecture proposal for 5 major application areas of a digital twin-based smart city	The study remains as basic research	No
Pan et al., 2022	The study proposes deep learning-based BIM pipeline using laser scanning and photogrammetry technology	So far, only simple geometric shapes can be extracted	No
Pantoja-Rosero et al., 2022	The study creates structure from motion and deep learning-based LOD2 and LOD3 models	Only facades and openings of buildings can be extracted	No
Otero et al., 2022	The study creates a roof BIM model in gbXML format using point cloud and machine learning	Only roof models can be created	No
Naderi and Shojaei, 2023	The paper reviews various infrastructure digital twins architectures and twinning technologies and interoperability solutions	The study remains as a basic research	No

some studies sought to reduce the geometry information of CAD models (3D models) only (Fang et al., 2021). Due to omitting various property information, research on service provision at the smart city level can be considered insufficient.

The digital twin system at the city level is an urban operation and management system to manage various services in all spatial ranges and thus requires spatial information of geography and facilities. There have been several studies using LIDAR point clouds to construct spatial information at the city level for urban objects such as geographical features, buildings, roads, bridges, and tunnels, but most of them focused on visualization with increased accuracy (Xue et al., 2020). As such, a methodology for real-time visualization in a small range with the use of point clouds has been proposed. However, since it is also a method of collecting and processing large amounts of point cloud data, real-time visualization at the city level has a limitation in data processing speed, and there is also a limit in that only geometry information can be visualized. This suggests that although line and face generation procedures are undergone, additional work is required to convert them into urban objects with property information for the provision of various Smart City services.

Accordingly, information on GIS and BIM that contain geometric and property information are necessary and many efforts have been made to integrate them for applications in smart city fields,

such as facility management, energy management, and disaster management (Gröger and Plümer, 2012; Mignard and Nicolle, 2014; Kang and Hong, 2015; Niu et al., 2015; “ISO - ISO/TS 19166:2021 - Geographic information — BIM to GIS conceptual mapping (B2GM)”); Lu et al., 2020; Schrotter and Hürzeler, 2020; Deng et al., 2021; Karimi and Iordanova, 2021; Shahat et al., 2021). However, according to 76 research cases related to BIM-GIS integration between 2008 and 2018, GIS and BIM models are configured based on different schemas, which poses difficulties in the complete conversion of the two standards (Wang et al., 2019), and more efforts are needed. Even if BIM and GIS information integration is done, it is difficult to implement all smart city services with CityGML's surface object, so BIM information, a solid object, should be utilized together. And the size of the BIM model of one structure often ranges from several hundred megabytes to several hundred gigabytes depending on the characteristics and size of the building, leaving another task of utilizing city-level information in real-time (Dou et al., 2020). In addition, since BIM, which contains all the information that can explain the entire building in detail, must be managed and visualized in real-time, it can cause too low or floating update rate during real-time rendering, making it inconvenient for users (Johansson et al., 2015), so lightening of BIM is essential.



*LOD 4: Room object, **LOD 300: Building element(wall, slab, column, room, etc.) object

Fig. 1. Digital Twin Development Type

2.2 Digital Twin System Cases

Lee et al. analyzed 65 studies related to the construction of smart city digital twins between 2008 and 2020 and examined the ratio of spatial information types used in each research. The results show that although research regarding CityGML and IFC integration occupied the largest proportion, most of the research was at the level of integration in the data model phase. In addition, it was done on the scale of a small community composed of one or several buildings for testing, and there was no case in which a city-level digital twin system was built (Lee et al., 2021).

Recently, some private development corporations such as Bentley, Dassault Systemes, PlaceTech, All4land, Gaia3D, and EGIS have built city-level digital twins and provided various services on a trial basis. Bentley developed the iTwin platform (Bentley Systems, 2022) to provide a foundation for building a digital twin solution. Dassault Systemes implemented the entire country of Singapore in three dimensions and developed Virtual Singapore (Dassault Systems, 2022) to facilitate policy decision-making processes, and PlaceTech built a VU.CITY (VU.CITY team, 2021) platform to organize, visualize and analyze city data. In South Korea, an early-stage digital twin system demo was created such as All4land's S-Map ("S-MAP") and Gaia3D's Mago3D ("mago3D"). S-Map provides 3D maps, living information, real estate information, and environmental analysis by year (All4land, 2022), while Mago3D offers 3D object libraries such as 3D building shape, street trees, streetlights and bridges, weather simulation, hydraulic/hydrology/water quality simulation, landslide simulation and air quality data display functions (Gaia3D, 2022).

According to the above cases, the development types of digital twins can be categorized by classifying the digital twin model construction method by LOD, general property and physical property, and the environment for service development and provision (Fig. 1). Type 1 uses spatial information (indoor survey data, building outdoor images, etc.) provided by the government; Type 2 uses direct floor survey data (aerial survey data, indoor survey data, etc.); Type 3 uses CityGML data; Type 4 uses IFC data and CityGML data; Type 5 uses private enterprise solutions. All the types are used by directly inputting geometry and property information depending on the data type according to the model's LOD, or by extracting from BIM/GIS data. In the case of national spatial data, LOD1 models can be created by extruding, and LOD2, LOD3, and LOD4 models can be created by adding building exterior images and indoor survey data using 3D modeling tools. In addition, it is possible to create a LOD2.5 model directly with aerial survey data or to create a higher LOD model through additional processing. If there is a CityGML or BIM model, a geometric model covering all LODs can be extracted and used. Depending on the use case, web based 3D GIS engine or game engine can be used, depending on the use case service, a physics engine capable of physics simulation may be required. However, the development method of systems provided by these private corporations is not open to the public and requires expensive licensing fees (White et al., 2021). Also, most cases are in the initial stage of trying to develop services by recognizing 3D objects

and performing physics simulations such as fluid simulations, and a computer with very high specifications is required to handle a vast number of urban objects at the city level in real-time. Therefore, this study seeks to find a method for constructing a digital twin at the city level and a lightweighting method for the digital twin to provide real-time services.

3. Research Framework

This study aims to create a city-level digital twin information model for the provision of various services. To this end, a series of lightweighting processes such as data classification, lightweighting geometry information, defining LOD of information, matching information, and dynamic load method are proposed. Next, according to the lightweighting process, GIS and BIM data are used to make the digital twin model lightweight, and the service display method of energy use is presented as an example though lightweighting that utilizes a dynamic load in terms of system operation. Lastly, the digital twin lightweight method proposed in this study is verified via a capacity comparison between the existing model and the lightweight model and a comparison of frames per second (FPS) in terms of system operation, and the limitations of existing technologies and future research directions for digital twin system construction are proposed through discussions (Fig. 2).

This study used type 1 and type 4 from the Digital Twin development types mentioned in Chapter 2. As a development environment to implement a digital twin that can provide all the services of a smart city, a game engine was used, which includes a physical engine for physical law calculation and a real-time 3D visualization. In general, a game engine refers to a software development environment designed for the development of interactive application programs with real-time graphics visualization functions, such as video games. The main functions of the game engine include 1) real-time 3D graphics visualization using a rendering engine, 2) approximate physical laws calculation using a physics engine, 3) optimization and computational efficiency through multi-threading and memory management, and 4) ease of functional extension with a high degree of development freedom such as support for various languages and data formats (Lee et al., 2021). In particular, the game engine is also used for experiments on virtual city content because it is

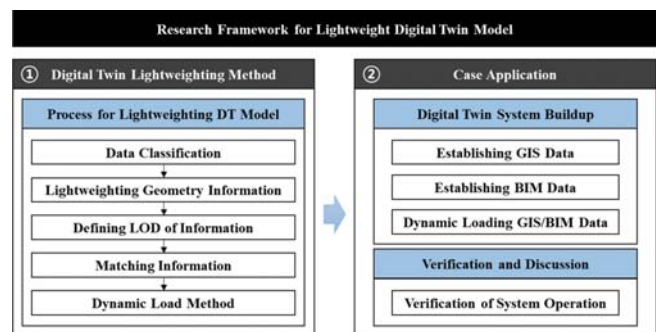


Fig. 2. Research Framework

easy to link with HMD (Head Mount Display) for VR (Virtual Reality) content production and experience (Mohammadi and Taylor, 2020). Moreover, it is suitable as a tool for smart city digital twin system development because it is also used in GIS data-based physics simulation studies (Harrap et al., 2019).

4. Digital Twin Lightweighting Method

This chapter presents the five steps of the method as in the process shown in ① of Fig. 2. These are step 1: classifying GIS and BIM data according to the purpose; step 2: GIS, BIM geometry information lightweighting; step 3: defining LOD information of Digital Twin for service display; step 4: matching geometry and property information; and step 5: dynamic load method for lightweighting.

4.1 Data Classification

GIS shape data provided at the national level contains various types of information according to LOD. All GIS data have geometry information from a macroscopic perspective and various property information such as building site address, use, name, height, and the number of floors. The geometry information is classified into map information with LOD 0, building shape information with LOD 1, and shape information including roof details with LOD 2, while the property information is classified by range, such as city, district, site, and building.

Like GIS data, BIM data also has geometry information and property information and contains detailed and diverse information for each building element from a microscopic point of view. Each IFC for buildings is a text-based file, and it requires a long loading time because many string processing operations are performed in the parsing stage for visualization. Therefore, to implement a digital twin at the city level, data should be classified by range, such as building, storey, space, and element, and selectively extracted and used if necessary.

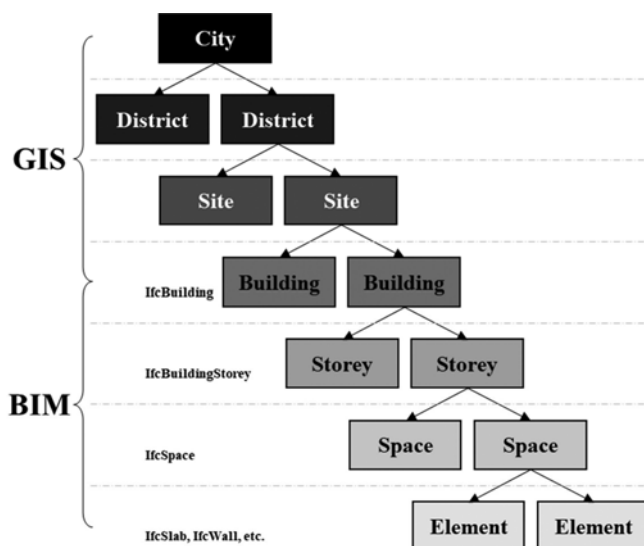


Fig. 3. Building Information Hierarchy

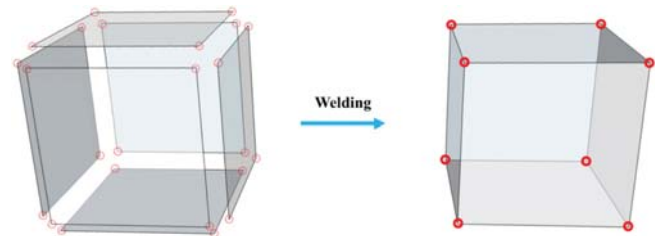


Fig. 4. Welding Concept

Figure 3 shows an example of a hierarchy for the organization of GIS and BIM data, in which data is structured in macroscopic and microscopic aspects to help readers understand the process. Both GIS and BIM contain overlapping information entities such as building name, longitude, latitude, and address. So, GIS and BIM can be linked using those entities, and they can be selectively extracted and used as necessary. In addition, various data such as real-time data collected from sensors and legacy data for each service are used in the digital twin, and tables should be added and managed in the database according to the use case.

4.2 Lightweighting Geometry Information

4.2.1 GIS Geometry

For GIS 3D modeling, there is a method by which a footprint of a building, the geometry information of the shape file, is extruded using the height information in the vertical direction and grouped as a building object. However, since this method requires height information of the building, it is difficult to create an accurate model from the existing shape files. Accordingly, a method of automatic 3D modeling using LiDAR point clouds to establish shape data has been proposed in recent years. However, it is not an object model as shown in Fig. 4(left), and the number of vertices of each building shape is more than necessary, because the surface and the line are separated. Therefore, by removing a large number of unnecessary vertices and edges in the same position on the geometry via welding as shown in Fig. 4(right), the shape can be represented using the minimum number of vertices and edges. It can be confirmed that when the welding operation is performed in one building model as shown in the example in Fig. 4, the number of vertices is reduced by one-third, and it is thus possible to realize significant progress toward lightweighting when implemented at the city level.

4.2.2 BIM Geometry

The performance of the most commonly used file formats for reducing the weight of files containing geometry information in BIM information, such as obj, glb, and dae, was compared using a BIM model of a general apartment with more than 20 floors and five buildings. The comparison revealed that there was no object and vertex loss in all three types of file formats as shown in Table 2, and the comprehensive evaluation found that the GLB file format showed the best performance in terms of file size and loading speed.

In addition, as shown in Table 3, the number of vertices can

Table 2. Performance Comparison (obj versus glb versus dae)

File Type	Object	File Size (MB)	Object (Unity)	Vertex (Unity)	Loading Speed (sec.)	Performance				
						Size Efficiency	Loading Speed	External Dynamic Load	Vertex Loss	Comprehensive Evaluation
.obj	O	53.5	21,547	603,824	50	2	3	O	X	2
.glb	O	32.6	21,547	603,824	8	1	2	O	X	1
.dae	O	75.9	21,547	603,824	X	3	X	X	X	3

Table 3. Performance Comparison (glb versus weld glb)

File Type	Lightweighting	Object	File Size (MB)	Object (Unity)	Vertex (Unity)	Loading Speed (sec.)	Efficiency	
							Size	No. of Vertex
.glb	General	O	32.6	21,547	603,824	8	43%	67%
	Weld	O	18.6	21,547	201,277	5	Less.	Less.

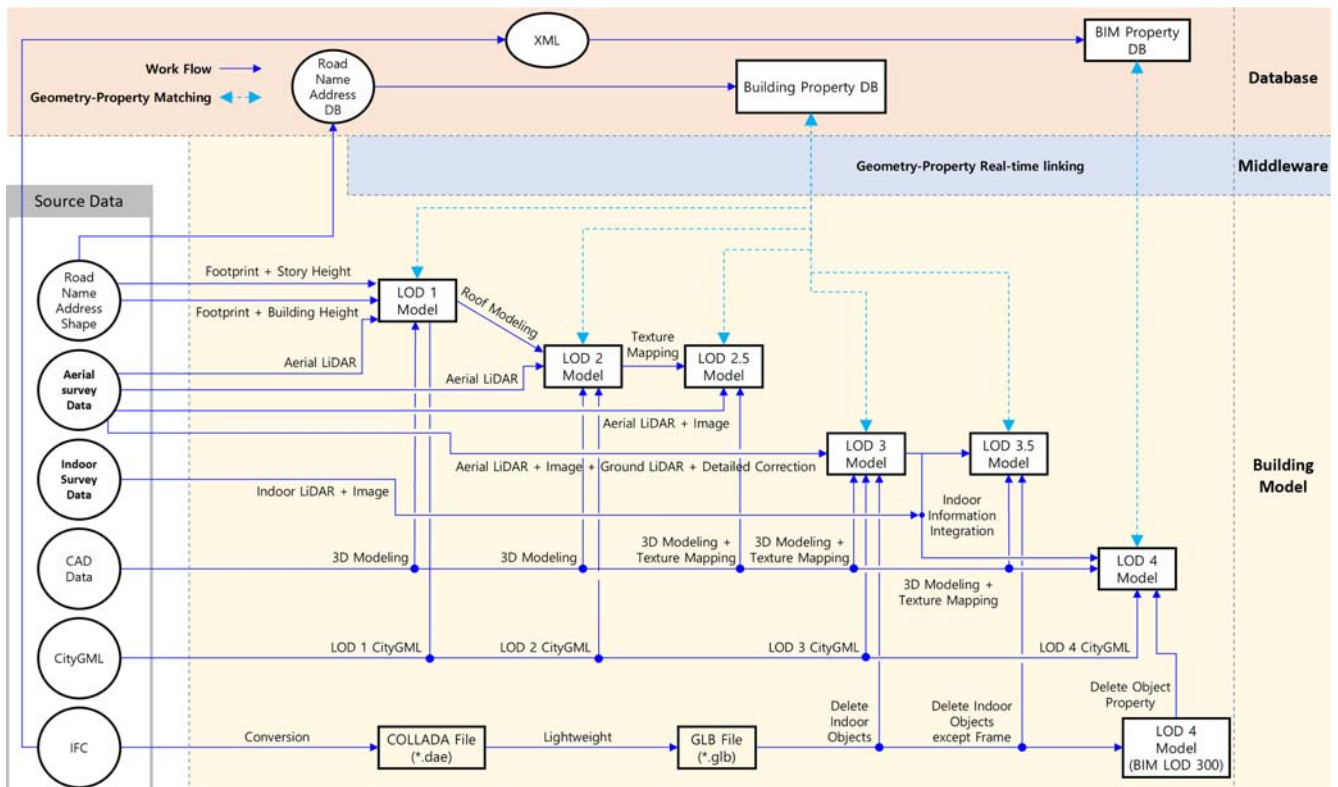
be reduced using welding operations in Unity to reduce the weight of the GLB file as in the case of the GIS shape file. Therefore, in this study, it is recommended to use the welded file in GLB format for the geometry information of BIM.

In addition, because it can be selectively extracted and used for each element of the building depending on the needs of each service, lightweighting can be realized via the simplification of the BIM model, except for unnecessary parts.

4.3 Defining LOD of Information

The concept of Level of Service (LoS) is needed to define the

LOD information of the digital twin for information display for each service. LoS Analysis is a major component that has a significant impact on asset management planning, and its purpose is to seek a balance between the service level and the cost that incurred from providing the service (MFOA, 2021). This study proposes to set the LoS according to the user requirements and define the LOD information of the digital twin. In this regard, there is a need to define the digital twin information required for service display in terms of the range, such as city, district, site, building, storey, space, and element according to use case characteristics and user requirements. In addition, the developer

**Fig. 5.** Data Modeling Method

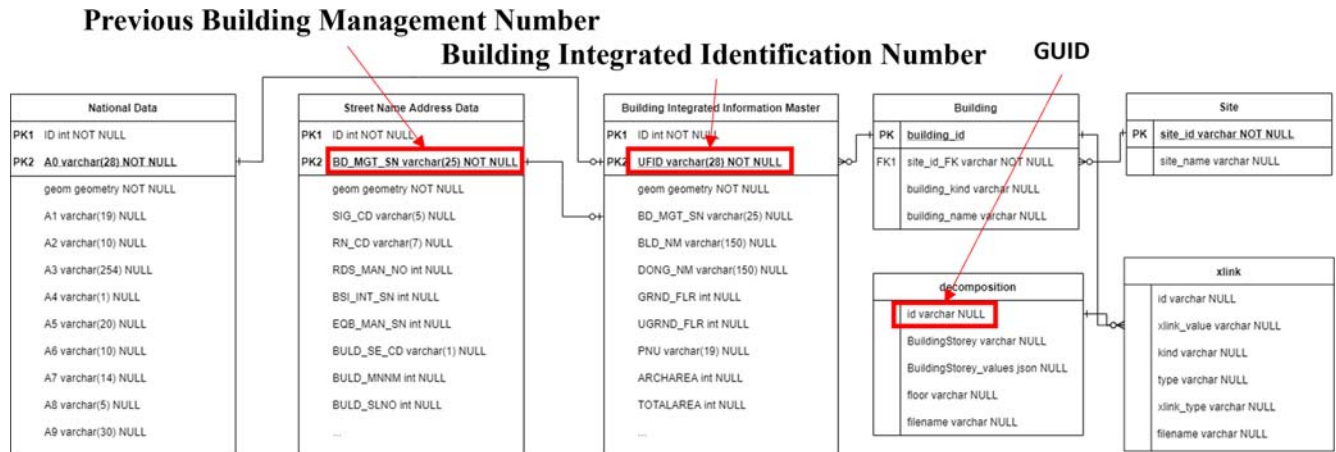


Fig. 6. Example of Entity Relationship Diagram

should establish LoS according to the requirements of stakeholders, who are provided with services (such as public servants, citizens, service providers, developers, etc.), and also define the LOD information of the digital twin accordingly.

In general, the LOD of digital twin model data required for each service is different from each other, and it should be developed so that the digital twin model can be utilized according to the minimum LOD required for each service. Fig. 5 shows the source data (road name address shape, aerial survey data, indoor survey data, CAD data, CityGML data, IFC data, etc.) used according to the LOD of the digital twin model, workflow, and the connection of required geometry and property data is shown. For example, the LOD 1 model is modeled by creating a surface with the footprint and extruding it by applying the story height or building height; the LOD 2 model is modeled by roof modeling in addition to the LOD 1 model; the LOD 3 model is modeled using aerial LiDAR data, related images, ground LiDAR data, and by detailed correction; the LOD 4 model can be modeled by extracting CityGML or IFC data. In this way, depending on the use case, you can select the utilization and modeling method of modeling data, this study defines the LOD 2.5 model, which added texture mapping similar to the actual object in the real world to the exterior of the LOD 2 model to improve the realism in detail, and the LOD 3.5 model, which can check the geometry (including interior main members without property information) of the interior space without increasing the system load. However, to provide the optimal digital twin service, it is necessary to divide the LOD in more detail according to the characteristics of the use case and the user's requirements and to select and utilize the LOD according to the defined LoS. For this, we can refer to a segmentation method such as the 16 LOD sets presented to overcome the disadvantages of the existing CityGML (Lehner and Dorffner, 2020).

4.4 Matching Information

To operate the Digital Twin System, city object information in the real world and information for each use case collected in various ways should be linked and integrated into one system.

For matching GIS shape information as shown in Entity Relationship Diagram (ERD) in Fig. 6, information IDs matched for each shape file such as 'Previous Building Management Number' and 'Building Integrated Identification Number' are needed, and the GUID of the BIM object is required for matching information for each BIM IFC object. To match GIS and BIM, there must be an entity with the same value, such as building name and address, which can be written in the tool for BIM. In this way, information for each service and range can be utilized and displayed by linking GIS and BIM information that can be matched with each other, and further matching real-time or legacy information for each use case.

4.5 Dynamic Load Method

The dynamic load method proposed in this study is to load the necessary data into the memory in real time from the system for operation. This way, lightweighting in terms of the system operation can be realized by selectively extracting, loading, and visualizing massive data from the Digital Twin server (explained in section 5.1) as needed.

The Digital Twin model is implemented in virtual space based on GIS data, and the model proposed in this study is placed based on the UTM-K coordinate system in which the point of intersection between 127.5 degrees longitude and 38 degrees latitude is set as the reference origin. In addition, each facility object has its own coordinates with respect to the UTM-K coordinate system. Therefore, it is possible to implement a dynamic load technology that selects only the facility objects necessary for visualization according to the user's point of view and loads them into the system in real-time through the calculation of coordinates.

In the Digital Twin service, the user's viewpoint is the center point of the camera object placed with the Digital Twin models in the 3D virtual space, and the center point coordinate of the camera object can be recorded in the memory of the PC in real-time. The center coordinates of the Digital Twin models are stored in the DB of the Digital Twin server when the distance between the center point coordinates of the camera is recorded in

real-time and the center coordinates of the Digital Twin models of the facility can be calculated using vector operations of spatial coordinates. Thus, by the distance set by the developer, facility objects within the specific range from the center point coordinate of the camera are selected and visualized. In addition, if the number of facility objects increases depending on the implementation range of Digital Twin, the calculation of each facility location also imposes a significant performance load. In this case, a method of binding a specific range of geography and facility models together for tiling can also be a good lightweighting strategy. Fig. 7 shows factors considered when dynamic loading is carried out by creating a digital twin model of a region corresponding to an area of about 98 km² with 256 tiles. The distance between the center point coordinates of each tile and the camera center point coordinates is calculated every 0.3 seconds, and dynamic loading of tiles that are present within the reference distance is done. When the user intends to cancel loading while the tile is being

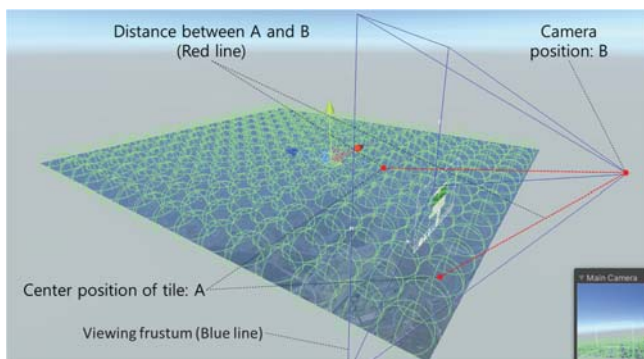


Fig. 7. Factors of Dynamic Load Method

loaded, the user determines whether the tile exists in the viewing frustum to execute cancelation. In this study, the reference distance was set to 2 km in consideration of performance. However, given the visibility range of the human body, it is desirable to set it to 2 km or more. As such, the memory load can be reduced by making the digital twin model lightweight in a way the necessary model, not the entire city model, is loaded in real-time.

5. Case Study

This chapter examines the effectiveness of the process and the method to build a lightweight Digital Twin system presented above. To do this, a digital twin model with an area of about 50,000,000 m² was constructed using GIS shape data on four 00 neighborhoods (dong), 00 City provided by the government, and IFC files of 20 buildings in 00 apartment and 00 building at 00 University. In addition, an energy field that requires object information of LOD 4 was selected as a use case, and the results of energy usage predicted using BEMS (Building Energy Management System) data and open weather data were visualized by building a room for configuration.

For the case study, QGIS was used as a GIS data editing tool, PostgreSQL as a digital twin DB construction tool, 3ds MAX as a tool for 3D modeling modification, and Unity as an engine for visualization and simulation. In addition, a network module for communication between the Unity client and the digital twin server was developed using Node.js. As computers for testing the prototype operation of the developed digital twin system, one computer equipped with Intel I7-9700F CPU and RAM 32Gb, NVIDIA GeForce RTX 2070 SUPER GPU, and a server with

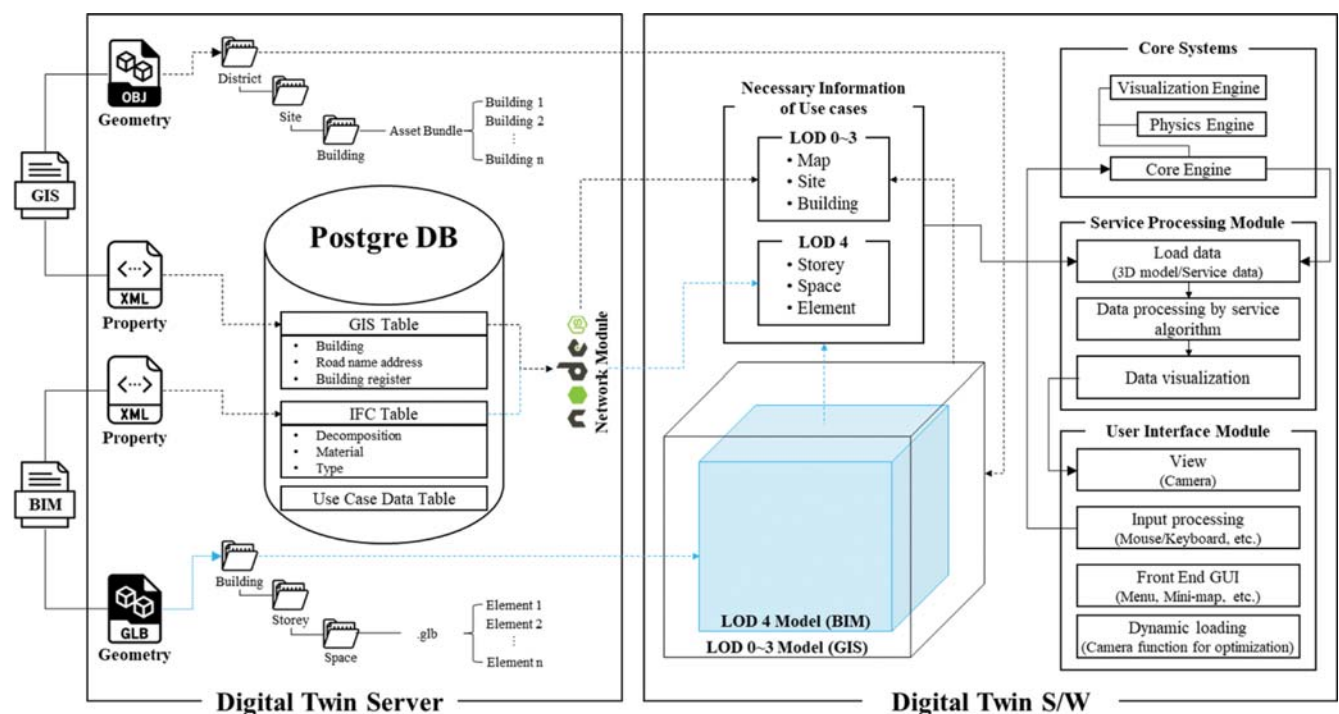


Fig. 8. Digital Twin System Architecture

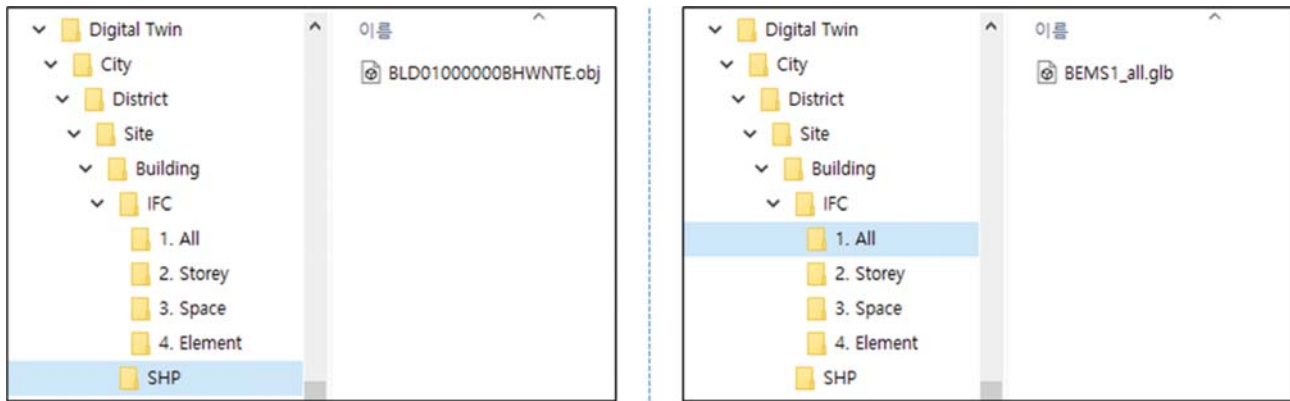


Fig. 9. Data Structuralization Concept

Intel I7-8700 CPU and RAM 16Gb were used.

5.1 Digital Twin System Buildup

As shown in Fig. 8, the architecture of the Digital Twin system built according to the lightweighting process proposed in this study consists of a Digital Twin server and Digital Twin software. The server has a Postgre database that stores the geometry information and the property information of the city object and use case-related data. All the information is provided to the Digital Twin software through a Node.js network module. The Digital Twin software comprises a core system that includes a visualization engine and a physics engine, a server processing module for implementing detailed functions for each service, and a user interface module to realize user operation-related functions. The core system is the most important part that determines the performance of the service, and it plays a central role in processing data input/output of the entire system and contributes to selectively visualizing the relevant results calculated in the service processing module according to the user requirements entered through the user interface module. As indicated by the dotted lines in Fig. 8, GIS and BIM data are split into geometry information and property information and loaded into the server. The geometry information is provided according to the required LOD through a query, and the property information is provided according to the required LOD via a node.js network module. The OBJ format was used for the geometry information of GIS, while the GLB format was used for the geometry information of BIM, and all property information was unified in XML format.

5.1.1 Data Classification

The Digital Twin system built in this study comprises GIS and BIM information, which is city object information, and energy use case information. For GIS information, the geometry information of the national core data shape file, road name address shape file, building integrated information master shape file, and 00 City LOD 2 shape file were classified by city, district, site, and building. For BIM information, the geometry information of IFC files of 00 building at 00 University and 20 buildings in 00 Apartment was classified by the storey, space, and element using

IfcConvert of IfcOpenShell. Fig. 9 shows an example of classifying and structuring the geometry information of GIS and BIM by range. The property information of GIS and BIM was stored in a separate property information table, and such information as energy use case-related BEMS data and national meteorological data was also stored in a separate use case table so as to display information through a linkage with city objects using GUIDs and unique IDs. The details are shown in ERD presented in 5.1.4.

5.1.2 Lightweighting Geometry Information

As mentioned earlier, the general GIS model is not in the form of a solid model based on parametric data, but it is saved as a surface model and is thus composed of countless vertices. Due to these objects in the form of a surface model, extra unnecessary load is applied in memory compared to the solid model when GIS data is visualized on the screen. To make the GIS model lightweight, the first welding operation was carried out to weld multiple overlapping vertices in the same location, and thus reduce the number of unnecessary vertices. In addition, in the LOD 2 model of 00 City built using LiDAR point clouds used in this study, some reverse faces exist as shown in Fig. 10(a), and although welding was performed in the 3D max program, the reverse face continues to exist as shown in Fig. 10(b). The reverse faces were corrected by creating a thickness on the model with the use of a 'shell' function in 3D max (Fig. 10(c)), and the results of the second welding operation confirmed that the number of vertices per building was reduced from 167 to 50 (Fig. 10(d)). In this way, data for more than 50 buildings has been made lightweight and uploaded onto the server.

The geometry model of the BIM file was extracted and processed in the GLB file format, and the file size was reduced by about 50%, and the number of vertices by about 65% as shown in Table 4.

5.1.3 Defining LOD of Information

This study proposed lightweighting methods for both GIS and BIM data, and an energy use case that requires LOD 4 was selected in the case study. For the energy use case, the service provision plan was applied to assume the user requirements in

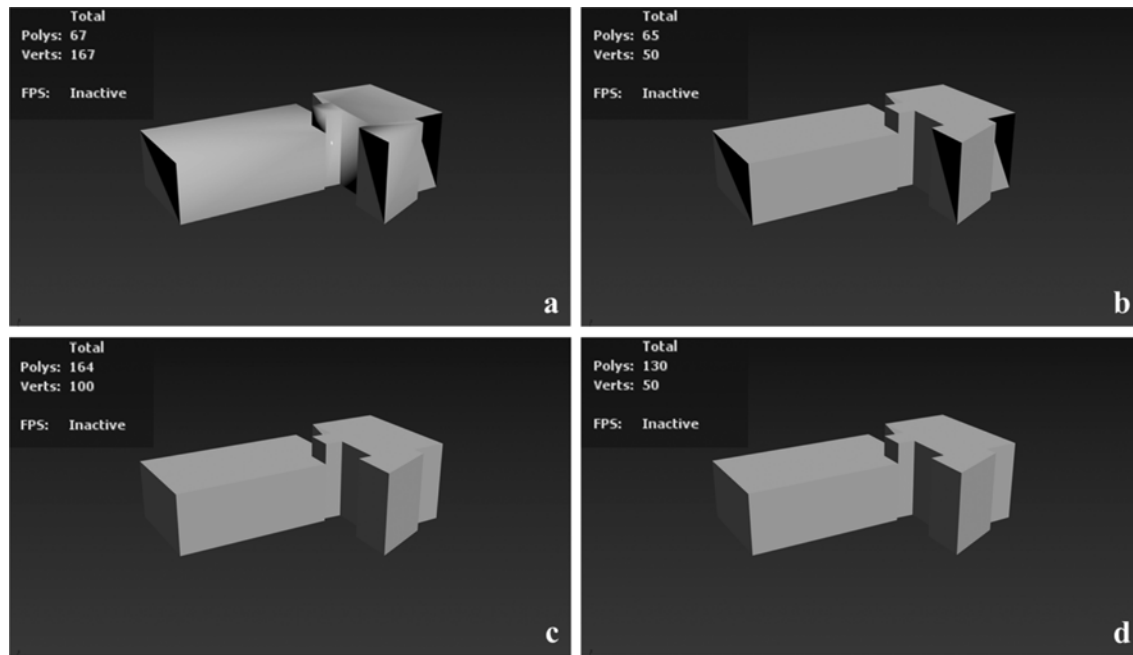


Fig. 10. GIS Geometry Data Lightweighting: (a) Before Welding, (b) 1st Welding, (c) Surface Processing, (d) 2nd Welding

Table 4. Lightweighting BIM Data by Welding

File Type	Light-weighting	Object	File Size (MB)	Object (Unity)	Vertex (Unity)	Loading Speed (sec.)	Efficiency	
							Size	No. of Vertex
.glb	Original	O	250.5	147,405	5,770,365	70	50%	65%
	After welding	O	124.9	147,405	1,999,260	47	Less.	Less.

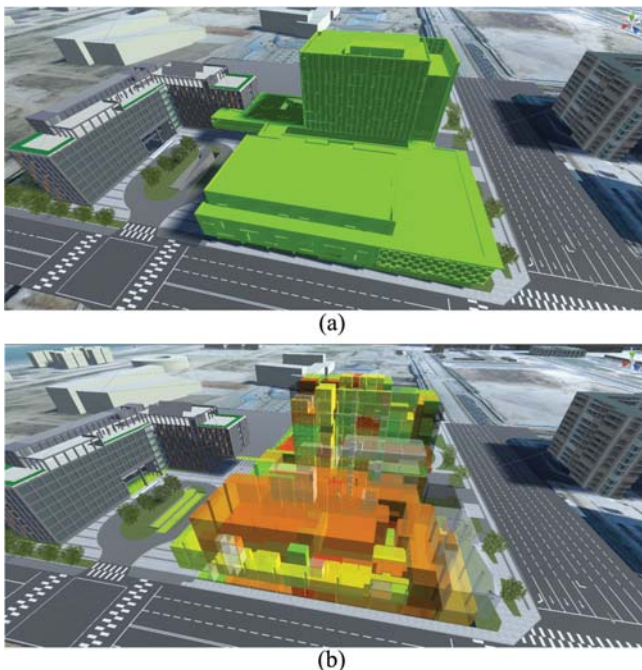


Fig. 11. Energy Service Visualization: (a) by Building, (b) by Room

the direction of displaying the energy usage results obtained by analyzing various energy-related data such as BEMS data and

open meteorological data. Services according to the service characteristics are displayed in units of buildings(a) and rooms(b) as shown in Fig. 11.

5.1.4 Matching Information

The data used in this study comprises open GIS data (national level), BIM data of the target building, and energy use case data. All data were connected by matching each data based on building information.

Figure 12 shows the ERD of energy use case Digital Twin DB. The digital twin DB is largely divided into GIS information, building information, BIM information, and energy use case information, and the data corresponding to each part is organized into tables for each purpose. With this ERD configuration, service display is possible as shown in Fig. 11.

The building information part consists of fixed tables: *PNU*, *Site*, *Building*, *BuildingHashMap*, *BuildingFloor*, and *BuildingRoomHashMap*, and it is used to link GIS information, BIM information, and energy use case information. The connection among GIS information, BIM information, and building information is possible by connecting the *PNU* data of the *Building total info._master* table of GIS information and *ifcSite* data among *ifcEntity* data of *decomposition* table of BIM information part with the *pnu* data and the *geom* data of the *PNU* table of the building

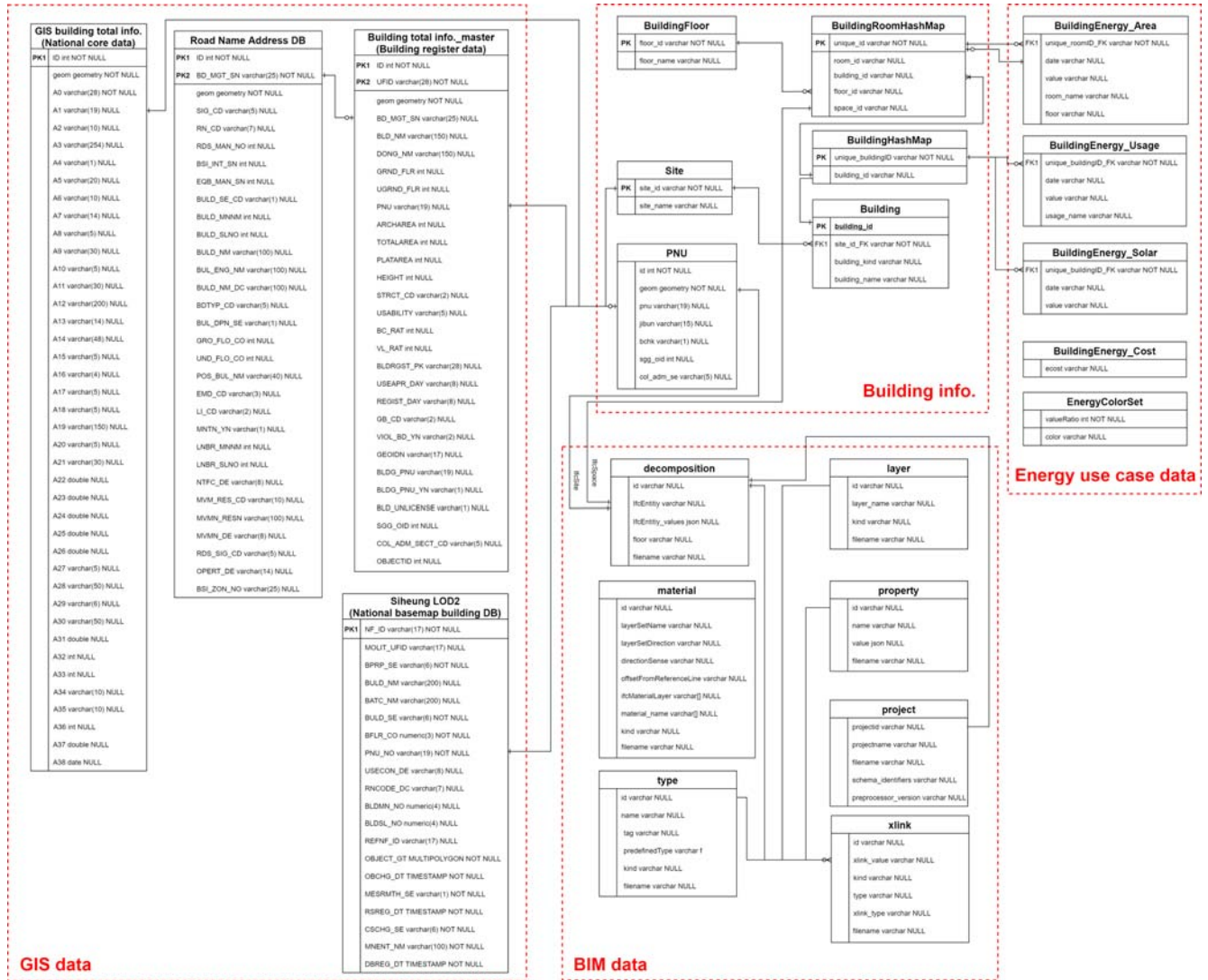


Fig. 12. Energy Use Case ERD

information part, respectively (Figs. 13, 14). *BuildingHashMap* and *BuildingRoomHashMap* tables are designed to implement unique IDs, and *building_id*, *floor_id*, and *space_id* are all integrated into *BuildingRoomHashMap*. Then, by matching the implemented unique ID with the *unique_roomID_FK* and *unique_buildingID_FK* of *BuildingEnergy_Area*, *BuildingEnergy_Usage*, and *BuildingEnergy_Solar*, it is possible to connect the energy use case part and building information (Fig. 15). GIS data are logically linked to building room data by connecting the *PNU* table to *Site*, *Building*, *BuildingHashMap*, *BuildingFloor*, and *BuildingRoomHashMap* tables with each ID, respectively, in regular series (Fig. 15).

The GIS information part consists of the *GIS building total info.* table, the *Road Name Address DB* table, the *Building total info._master* table, and the *Siheung LOD2* table of 00 City LOD 2 shape file. Some data in the tables may be partially omitted. The missing parts can be supplemented via the UFID (GIS building integrated identification number) data matching for the

national core data and the building integrated master data, and the *BD_MGT_SN* (road name address building management number, previous building management number) data matching for the building integrated information master data and the road name address DB data. In addition, for the GIS data of 00 City LOD 2 shape file, the missing part can be supplemented by matching it with the *pnu* data of the *Building total info._master* table (Fig. 13).

The table configuration of the BIM information part is the result of converting file based IFC information into ifcXML format, and the part consists of a total of seven tables: *decomposition*, *layer*, *material*, *property*, *type*, *project*, and *xlink*. When the IFC data is converted to ifcXML, the upper level is divided into header, units, properties, quantities, types, layers, materials, and decomposition, among which header, units, and quantity were excluded because they were unnecessary in the energy use case of this study. The *decomposition* table contains information about all elements that constitute the building. The *layer* table contains CAD layer

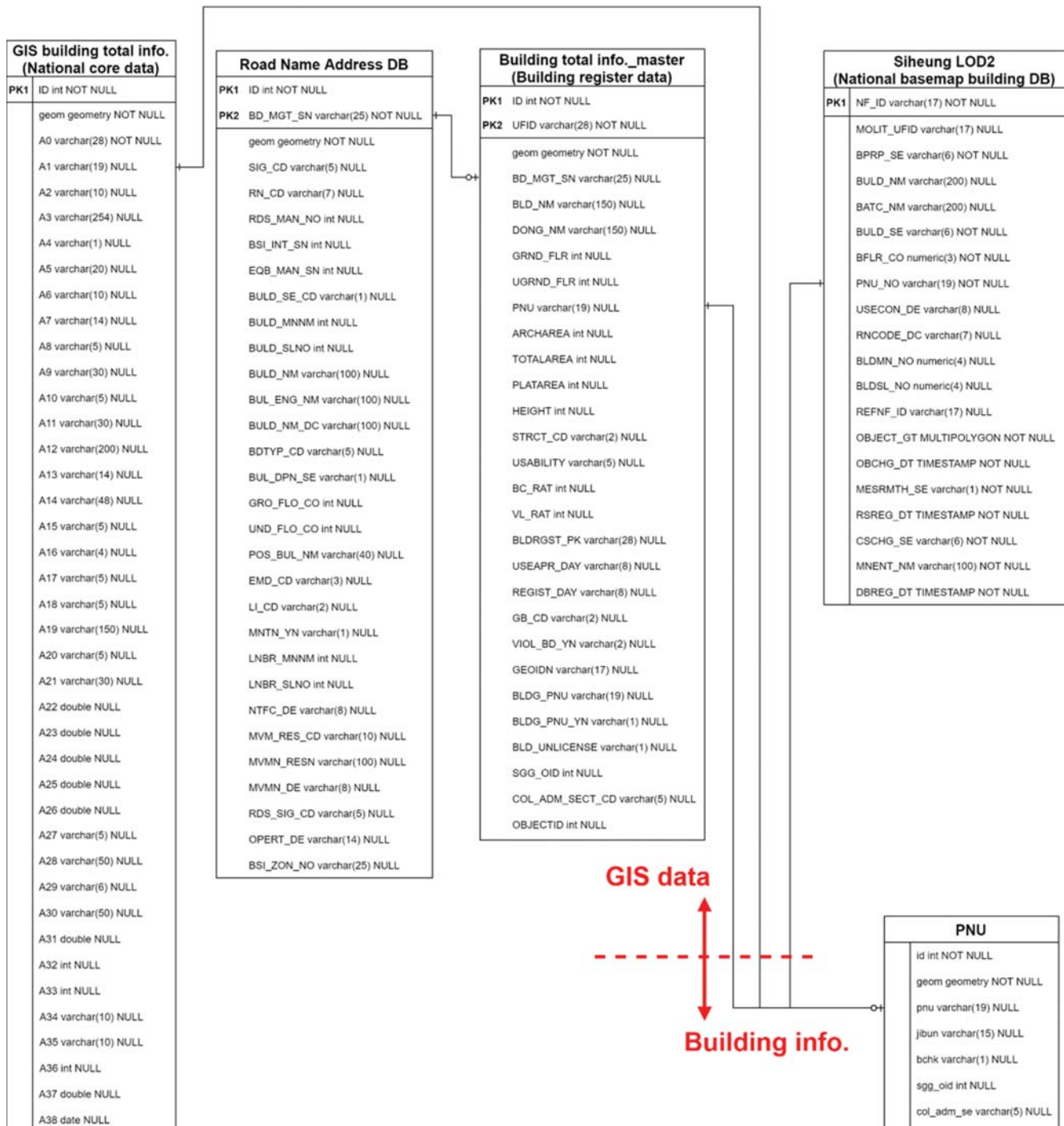


Fig. 13. A Part of Energy Use Case ERD: Match of the GIS Data and the Building Information (partial image)

information used for IFC data preparation. The *material* table contains material information of each element. The *property* table contains property information of each element. The *type* table contains type information of elements specified in a list of drawings, and the *project* table contains file name information of IFC data. Although the BIM information includes the details of the structural members of the building, part of the information is abbreviated due to the constraints in ERD. Consequently, much more information than that shown in Fig. 12 is included in the data.

For example, various data columns in addition to *ifcSite* and *ifcSpace* are included in the *ifcEntity* data of the *decomposition* table. For each piece of information in BIM, a cross-reference is provided to link the information to *xlink* data. Therefore, the *xlink* table is used to connect each table and thus link the reference data (Fig. 14).

The energy use case part consists of a total of five tables: *BuildingEnergy_Area*, *BuildingEnergy_Usage*, *BuildingEnergy_Cost*, *BuildingEnergy_Solar*, and *BuildingColorSet*. The

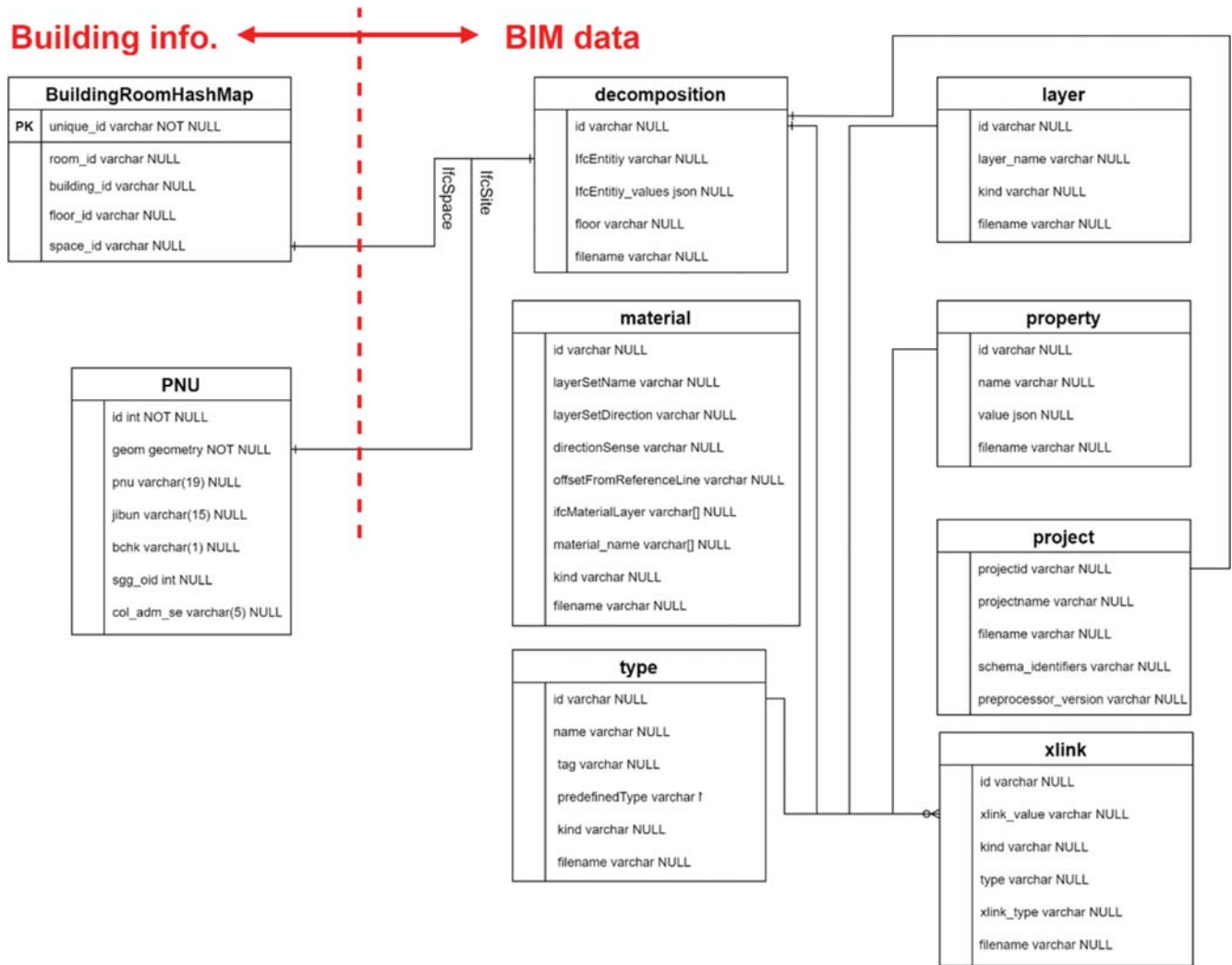


Fig. 14. A part of Energy Use Case ERD: Match of BIM Data and Building Information (partial image)

BuildingEnergy_Area table contains energy usage data for each room, and unique ID for each room is included for matching with the 3D model corresponding to each data. The *BuildingEnergy_Usage* table contains energy usage data for each building use, and the *BuildingEnergy_Solar* table contains data related to solar power generation in a building unit. The *BuildingEnergy_Cost* table contains additional information for calculating energy usage charges, while the *BuildingColorSet* is designed for color reference when visualizing energy usage through color change (Fig. 15).

5.1.5 Dynamic Load Method

The computer memory load was reduced by loading only the models that are present within 2 km from the user's field of vision, not the entire city model as tiles including buildings that exist within 2 km (arbitrarily determined by the author) from the camera center point coordinates are dynamically loaded as shown in Fig. 16 according to the dynamic loading method mentioned in Chapter 4.

5.2 Verification and Discussion

In this section, tests were done to verify the lightweighting methods. The test criteria include the number of vertices, file size, loading speed for step 2 (lightweighting geometry information) of the lightweighting method, FPS for step 1 (data classification), step 4 (matching information), and step 5 (dynamic load method) of the lightweighting method.

The GIS geometry information of about 50 building units showed a reduction in the number of vertices from 49,734 to 24,730 as shown in Fig. 17. For the geometry information of BIM, the file size was reduced by about 50%, and the number of vertices by about 65%, respectively, as shown in Table 4. In addition, since the objects (space) can be selectively visualized according to the service display method, a significant lightweighting effect was achieved as the file size was reduced by about 80%, the number of objects by about 84%, and the number of vertices by about 80% as shown in Table 5.

Lastly, FPS was compared with the existing method through a comparative test using IFC data for the entire building. In general, higher FPS values are required because lower FPS values pose

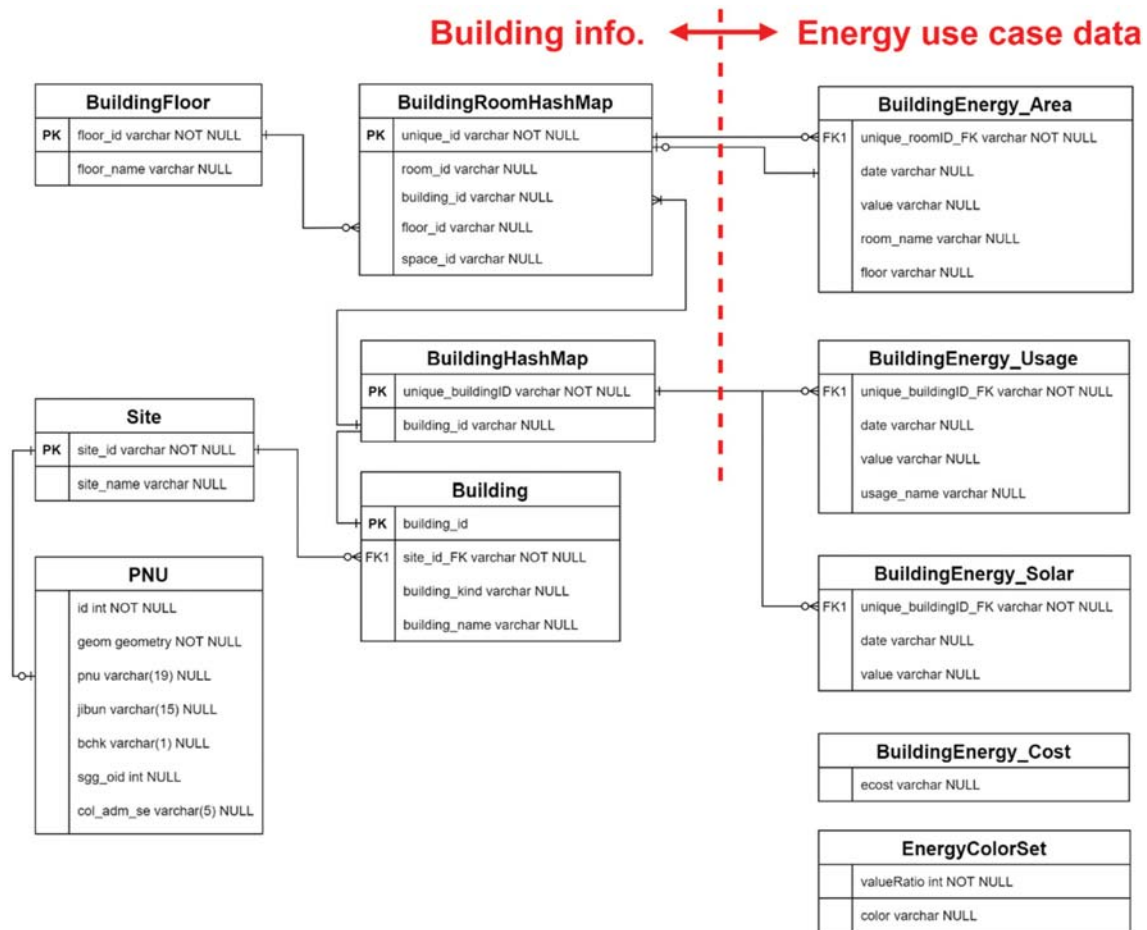


Fig. 15. A Part of Energy Use Case ERD: Match of Energy Use Case Data and Building Information



Fig. 16. Dynamic Load Capture

difficulties in observing the geometry information. The target scenario for the experiment is an energy usage visualization service, and the above-mentioned IFC data of 00 building at 00 University and 20 buildings in 00 Apartment were used as its targets. And a digital twin S/W prototype developed using the

Unity game engine was used in this study. To be specific, the digital twin model was implemented using the IFC data of the target buildings, and the energy usage for each building and floor was then visualized in a way that uses a method to identify the lightweighting effect between the method of using the entire IFC file and the method of using only ifcSpace (Fig. 18).

To compare the quantitative lightweighting effect, the FPS used to determine the expected performance of dynamic 3D graphic contents was measured to confirm the difference in the execution of the visualization service for each building and floor with the use of each method. The FPS had an error rate of about 10%, and the average of the values measured for five seconds was used as the result value. The experimental results are summarized in Table 6. The case of using only ifcSpace showed an improvement rate of about 968% compared to the case of using the entire IFC data.

The results of the above tests suggest that the lightweighting method proposed in this study has practical merits. However, since the shape data provided in the GIS field has different LOD, the lightweighting method may vary accordingly. For example, the model provided with LOD 0, which only has a footprint, only needs to be extruded as much as the height of the building and then objectified. However, the model provided with LOD2 modeled using LiDAR point clouds should undergo the welding

Table 5. Lightweighting BIM Data by Selective Extraction

Type	Object	File Size (MB)	Object (Unity)	Vertex (Unity)	Efficiency		
					Size	No. of Object	No. of Vertex
Original IFC	O	250.5	147,405	5,770,365	80% Less.	84% Less.	80% Less.
IfcSpace	O	50.23	23,632	1,163,876			

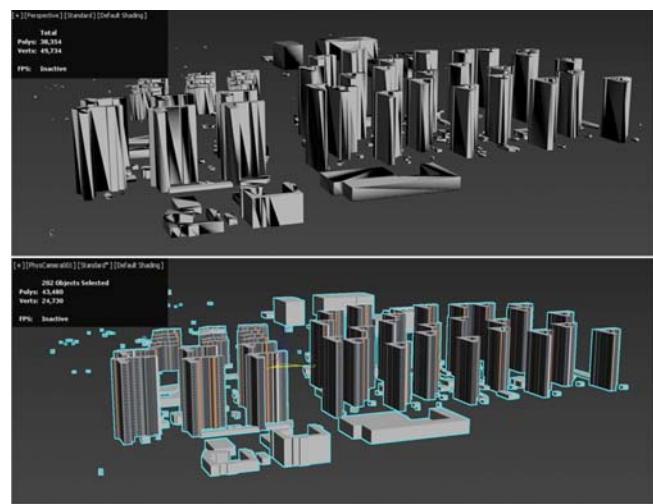


Fig. 17. GIS Data Lightweighting

Table 6. Performance Comparison of FPS

Type	FPS	Efficiency
Original IFC	2.2	968% improved
IfcSpace	23.5	

operation proposed in this study. In the case of shape data used in this study, since it was difficult to extract only the reverse face due to the presence of overlapping parts of the reverse face and each building in the geometric model, a separate “shell” operation to correct the reverse face by creating a thickness in the model and two welding operations were carried out. For this reason, the number of vertices decreased, but the number of polylines increased. Therefore, if an algorithm that automatically extracts and restores the reverse face (Charton et al., 2021) is used to address the above problem, it is possible to make the geometry information of GIS more lightweight. As shown in Fig. 5, the data required for modeling for each LOD is diverse, and the modeling method may vary depending on the available data. According to the lightweighting process presented in this study, modeling and lightweighting that encompasses the entire LOD can be performed, but it is necessary to apply more subdivided according to use cases.

In step 3 of the lightweighting processes, the digital twin model information required for energy-related use cases was applied by assuming the user requirements as an example. Therefore, if systematic and detailed guidelines such as a process that can set LoS (Level of Service) goal that reflects the user requirements related to service display for each use case are developed in the

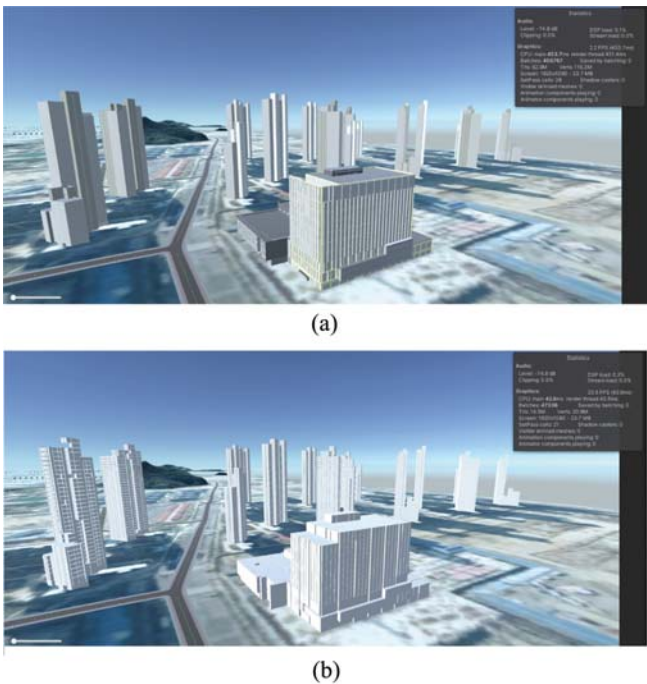


Fig. 18. BIM Data Lightweighting: (a) Entire IFC Model, (b) Selectively-extracted Model, IfcSpace

future, unnecessary elements can be filtered out according to the service requirements, thus reducing the load of real-time dynamic loading. In this study, the range for the dynamic loading of the building unit was arbitrarily limited to 2 km or less. It is necessary to designate a range in which the optimal performance can be implemented through multiple tests within various unit distances.

Lastly, if a method of visualization according to the required LOS through texture adjustment in terms of system operation mentioned in the introduction is devised to provide services, it will be possible to facilitate the provision of services depending on the specifications of the user’s device (i.e., PC, media board, tablet PC, etc.).

6. Conclusions

In relation to the development of an urban information model to implement a digital twin at the city level, many efforts have been made to integrate and utilize BIM-GIS data. However, there is a limit to practical applications because it is sometimes difficult to provide various real-time services in the era of a web environment due to problems with the capacity available for storing large amounts of data. Therefore, in this study, the digital twin construction type

was investigated through the existing case studies, and a lightweight digital twin construction method using a game engine (physical engine) has been presented by selecting a development type that can implement all services of a smart city.

For this, the BIM and GIS information lightweighting process has been presented as a method to provide various services that a smart city offers based on the digital twin, and the lightweighting method has also been proposed. The method according to the process comprises five steps: step 1: classifying GIS and BIM data according to the purpose, step 2: GIS, BIM geometry information lightweighting, step 3: defining LOD information of Digital Twin for service display, step 4: matching geometry, property and use case information, and step 5: dynamic load method using a selective real-time loading method.

In addition, a lightweight Digital Twin model was built based on real cases according to the above lightweighting process and the method. Through performance tests with indicators such as the number of vertices, file size, loading speed, and FPS, a possibility of service display in units of detailed elements rather than the existing units of buildings have been opened up, and its effectiveness has been verified in terms of system operation. The test results have confirmed that in the GIS geometry information, the number of vertices was reduced from 49,734 to 24,730, the file size by about 80%, the number of objects by about 84%, and the number of vertices by about 80%, respectively. In addition, the case of extracting only the IfcSpace entity showed an FPS improvement rate of about 968% compared to the case of using the entire IFC data.

In this study, a lightweight digital twin model with an area of 50,000,000 m² was built at the city level. Therefore, it is expected that the lightweighting method proposed in this study will serve as a basis for efficient city information processing and contribute to providing various smart city-related services in real-time in the web environment. However, further research regarding GIS shape data surface reverse processing and texture-related lightweight method, which were not addressed in this study, is needed to make the digital twin system more lightweight. In addition, depending on the data characteristics available, in addition to the LOD2.5 and LOD3.5 examples presented in this study, more detailed LOD definitions of roadway systems, utility lines, other facilities, etc. as well as buildings will be required. A smart city includes various services in various fields, and the requirements for each service characteristic are different, so when building a digital twin, it is important to set the LoS according to the service characteristics and set the detailed LOD of the digital twin model accordingly.

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